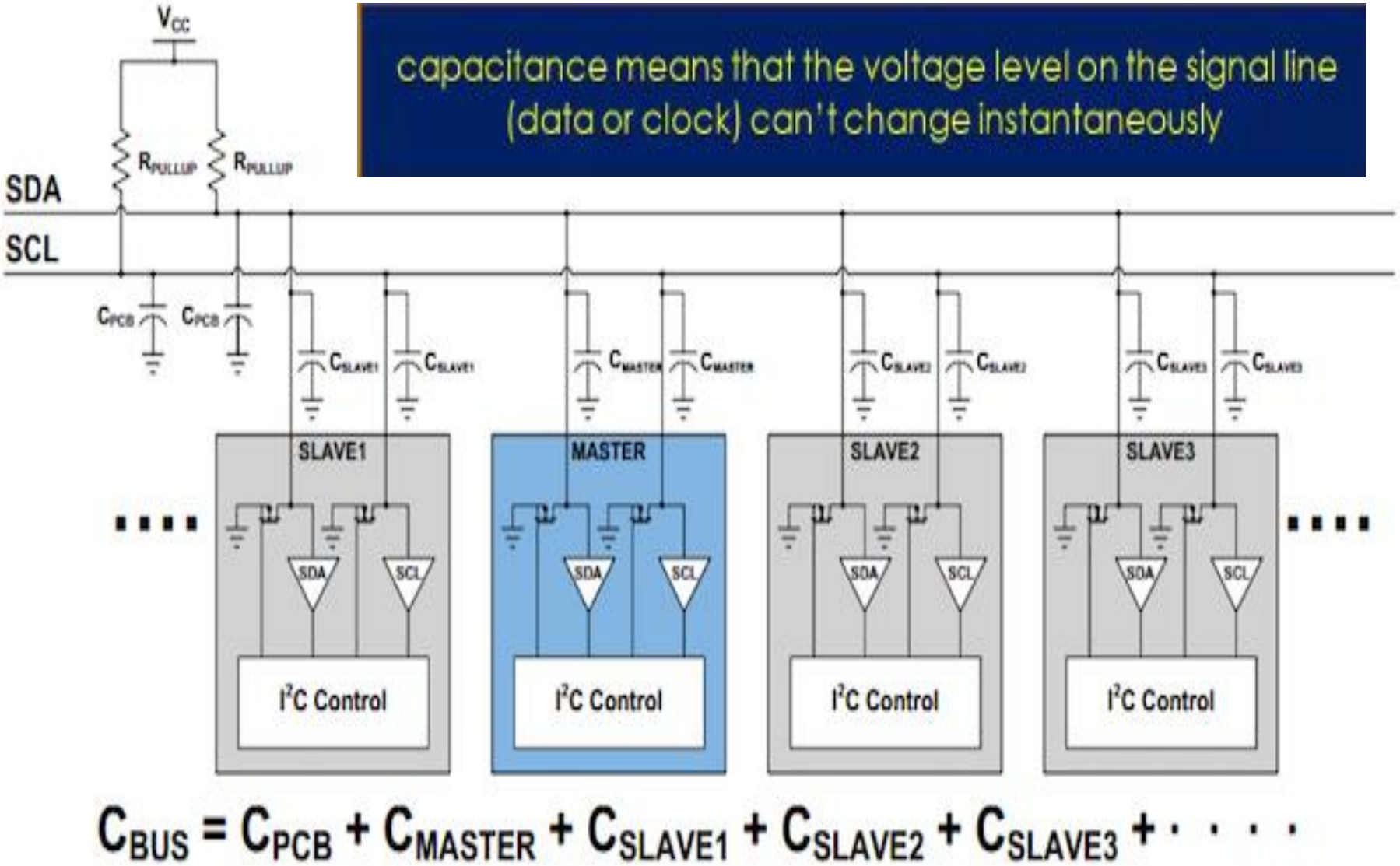
Typical I²C bus with associated bus capacitance.



Limitation to Number of slaves

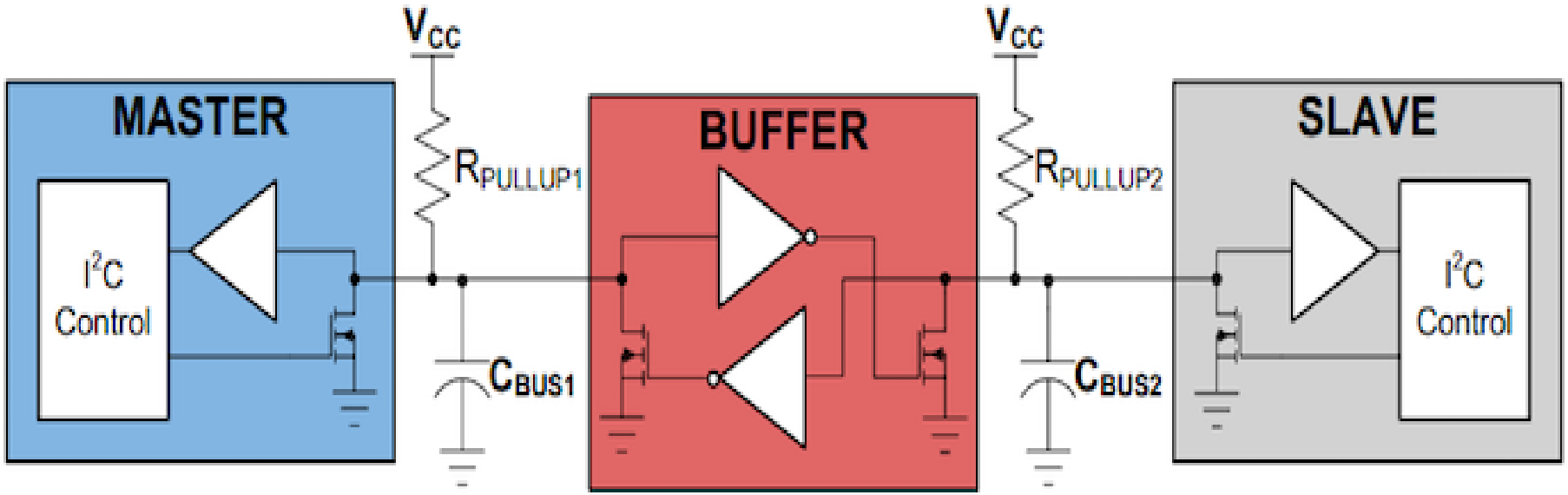
- Need to comply with the standard :

➢ When operating in Standard or Fast mode, the I^2C bus has a maximum bus capacitance of 400 pF. &

➢ In Fast Mode Plus, this is increased to 500 pF, no more devices can be placed on the bus.

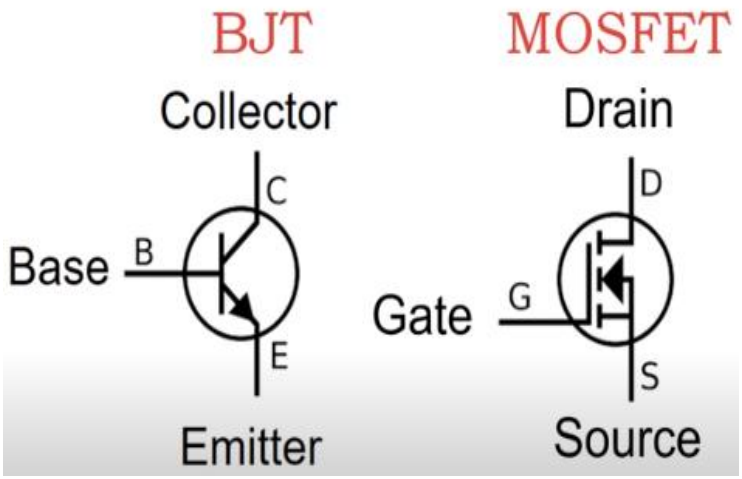
➢ Bus capacitance limits number of slaves ,and also total length of the bus.

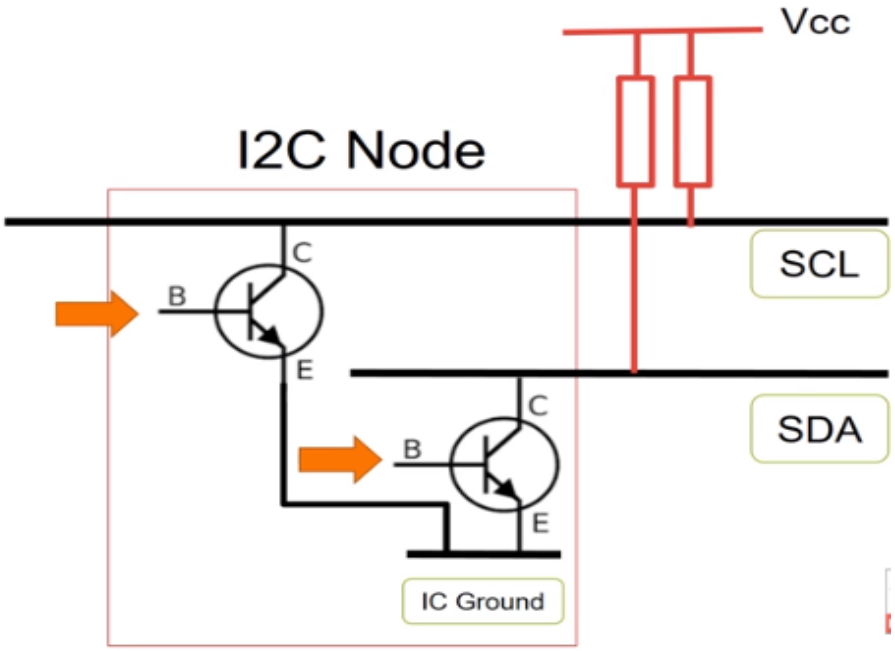
Buffer inserted into an I^2C bus, cutting bus capacitance in half

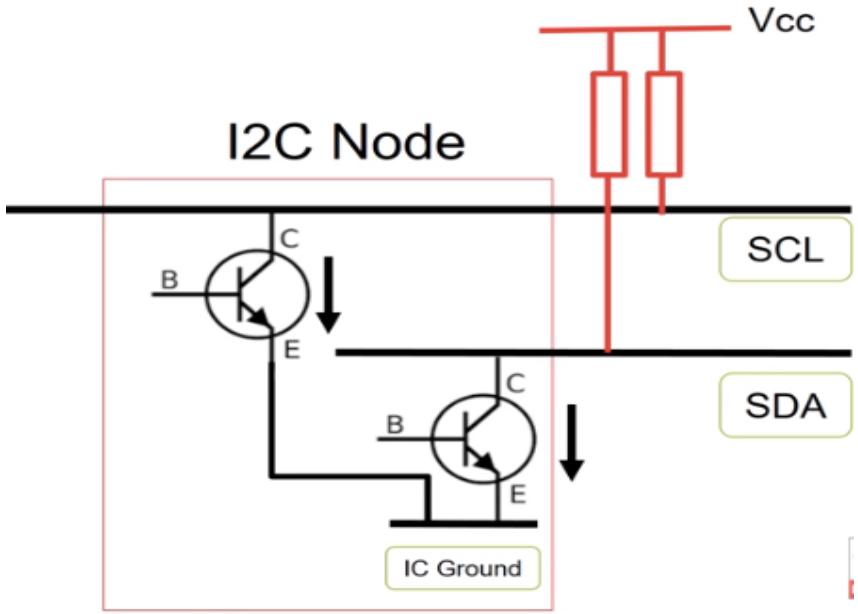


- One way of getting around this is to add an I^2C buffer to your design.
- The I^2C buffer divides the I^2C bus into two separate buses, while still allowing devices to communicate across it. This effectively cuts your total bus capacitance wherever the buffer is placed, because the separate buses have separate bus capacitances

•The I^2C bus drivers are "**open drain**", meaning that they can pull the corresponding signal line low, but cannot drive it high.







Pull-up Resistor(R_p) Calculation

R_p (min) is a function of VCC, VOL (max), and IOL:

$$R_P(\min) = \frac{(V_{CC} - V_{OL}(\max))}{I_{OL}}$$

V_{OL} = LOW-level output voltage

I_{OL} = LOW-level output current

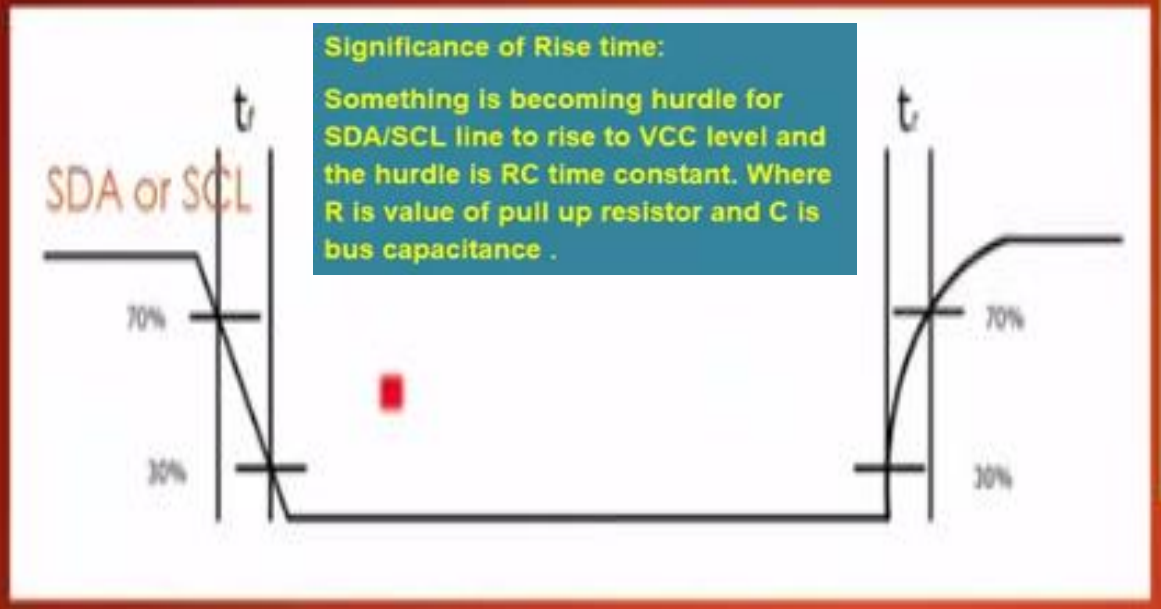
t_r = rise time of both SDA and SCL signals

C_b = capacitive load for each bus line

The maximum pullup resistance is a function of the maximum rise time (tr):

$$R_p(\max) = \frac{t_r}{(0.8473 \times C_b)}$$

Rise (t_r) and Fall (t_f) Times



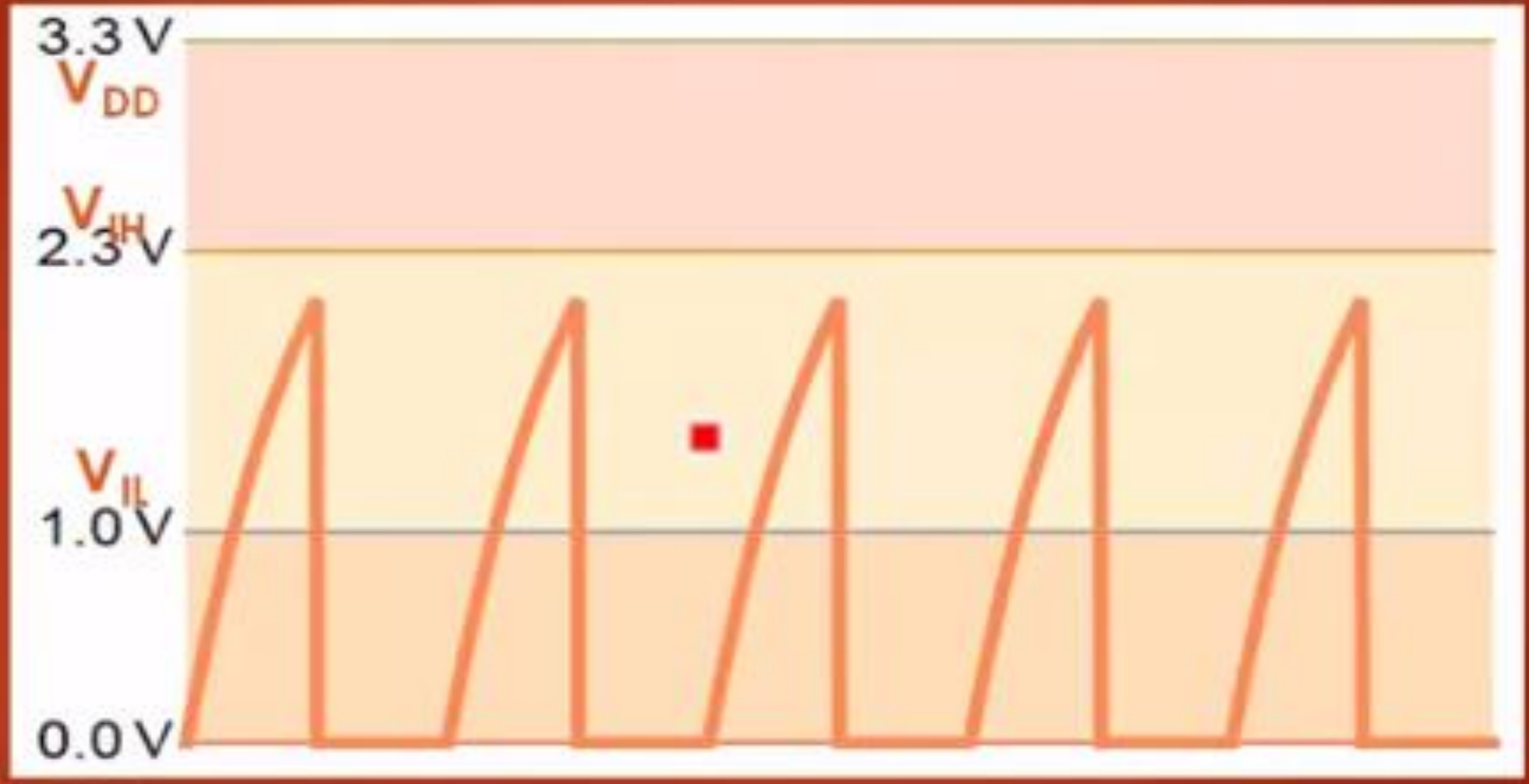
t_r is defined as the amount of time taken by the rising edge to reach 70% amplitude from 30% amplitude for either SDA and SCL

I2C spec cares about t_r value and you have to respect it while calculating the pull up resistor value.

Higher value of pull up resistors(weak pull-ups) increases t_r value. (not acceptable if t_r crosses max. limit mentioned in the spec)

Lower value of pull up resistors (strong pull-ups) decreases t_r value (good) but they also lead higher current consumption (bad)

Using very high value of pull up resistors may cause issues like this where the pin may not able to cross the V_{IH} limit (Input Level High Voltage)



PULL UP RESISTOR CALCULATION

- Refer document through url
- http://www.ti.com/lit/an/slva689/slva689.pdf?ts=1590645968608
- Refer tutorial through urls
- https://www.youtube.com/watch?v=sGZe0aJsqBQ&t=92s
- https://www.youtube.com/watch?v=CEPFWAPDy7o